



**SYDNEY BOYS HIGH  
SCHOOL**  
MOORE PARK, SURRY HILLS

**2008**

**HSC Task 1**

# Mathematics

## General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Start each **NEW** question in a separate answer booklet.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.

## Total Marks – 64

- Attempt questions 1-5.
- All answers are to be given in simplest exact form unless specified.

Examiner: *D.McQuillan*

Start on a separate answers sheet.

**Question 1 (13 marks)**

**Marks**

(a) Evaluate  $\log_{10} 12 - \log_{10} 2$  to 3 decimal places. 1

(b) Evaluate  $\sum_{n=2}^5 3n - 1$ . 1

(c) Find using the formula  $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$  find the distance between the line  $5x - 12y + 15 = 0$  and the point  $(-1, 3)$ . 1

(d) Find the domain and range of the following 6

(i)  $y = |2x - 6|$

(ii)  $y = \sqrt{9 - x}$

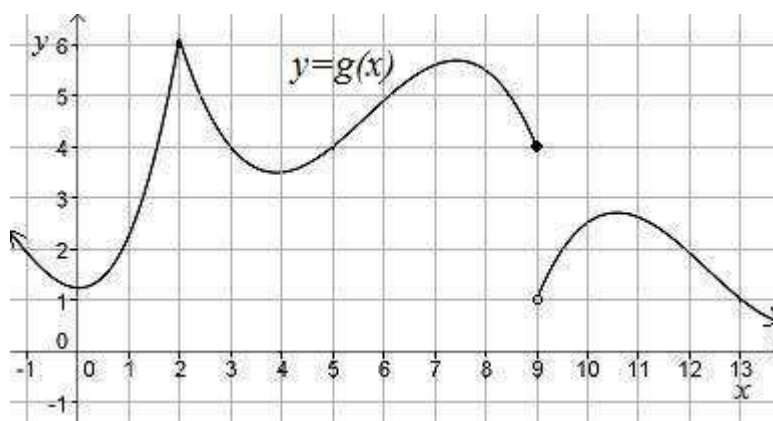
(iii)  $y = \frac{x}{|x|}$

(e) For the series  $2 + 5 + 8 + 11 + \dots$  2

(i) Find the value of the 20<sup>th</sup> term.

(ii) Find the sum of the first 20 terms.

(f) 2



(i) For what  $x$  value(s) is  $g(x)$  not continuous?

(ii) For what  $x$  value(s) is  $g(x)$  not differentiable?

**End of Question 1**

Start on a separate answers sheet.

**Question 2 (13 marks)**

**Marks**

- |       |                                                            |   |
|-------|------------------------------------------------------------|---|
| (a)   | Solve $3^x = 81$ for $x$ .                                 | 1 |
| (b)   | Show that $3x^2 - 4x + 2 = 0$ has no real roots.           | 1 |
| (c)   | Express $2x^2 - 5x + 8$ in the form $Ax(x - 1) + Bx + C$ . | 2 |
| (d)   | Differentiate the following                                | 5 |
| (i)   | $3x^3 - 9$                                                 |   |
| (ii)  | $\frac{x^2}{x + 1}$                                        |   |
| (iii) | $x\sqrt{x^2 + 1}$                                          |   |
| (e)   | For $f(x) = x^2 + 6x - 16$ find,                           | 4 |
| (i)   | the minimum value of $f(x)$ .                              |   |
| (ii)  | the values of $x$ such that $f(x) \geq 0$ .                |   |

**End of Question 2**

Start on a separate answers sheet.

**Question 3 (13 marks)**

**Marks**

- (a) Is the function  $f(x) = \frac{3x^3 + x - 1}{x}$  ODD, EVEN or NEITHER? 1
- (b) If  $\log_b 3 = 1.09$  and  $\log_b 4 = 1.38$  find 2
- (i)  $\log_b 12$
- (ii)  $\log_b \frac{2}{3}$
- (c) If  $f(x) = \frac{2x^2 + 5x - 3}{4x^2 + 16x + 12}$ , find the value of: 4
- (i)  $\lim_{x \rightarrow 0} f(x)$
- (ii)  $\lim_{x \rightarrow -3} f(x)$
- (iii)  $\lim_{x \rightarrow \infty} f(x)$
- (d) If  $\alpha$  and  $\beta$  are the roots of the equation  $2x^2 + 7x + 4 = 0$ , find the value of: 4
- (i)  $\alpha + \beta$
- (ii)  $\alpha\beta$
- (iii)  $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$
- (e) What is the equation of the parabola with vertex (1, 2) and directrix  $y = -1$ ? 2

**End of Question 3**

Start on a separate answers sheet.

**Question 4 (13 marks)**

**Marks**

- (a) Dean undertook to pay a charity \$200 one year, \$150 the next year, three-quarters of \$150 the third year and so on until he died. 4
- (i) What is the greatest sum of money the charity may expect from these donations?
- (ii) If Dean died after making 20 donations, by how much, to the nearest dollar, would the total received by the charity fall short of the maximum expectation?
- (b) Given that  $f(x) = x^2 - 6x$ . 3
- (i) Find  $f'(x)$  using differentiation by first principles.
- (ii) Hence show that  $2xf'(x) - f(2x) = 0$ .
- (c) Given the points  $O$ , the origin, and  $A(0, 2a)$ ; find and graph the locus of  $P$  such that  $OP \perp PA$ . 3
- (d) If  $\log_b a = p$  and  $c = a^2$ , find, in terms of  $p$ . 3
- (i)  $\log_b c$
- (ii)  $\log_c b$

**End of Question 4**

Start on a separate answers sheet.

| <b>Question 5 (12 marks)</b>                                                                                                                                                   | <b>Marks</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| (a) Sketch the graph of $y = \log_2(x + 2)$ showing all important features.                                                                                                    | 2            |
| (b) Sketch the following region.<br>$y \leq \sqrt{25 - x^2}$ $x + 2y - 5 < 0$                                                                                                  | 3            |
| (c) Each summer 10% of trees on a certain plantation die out, and each winter, workmen plant 100 new trees. At the end of winter 1990 there were 1200 trees in the plantation. | 7            |
| (i) How many living trees were there at the end of winter 1980?                                                                                                                |              |
| (ii) When will the plantation fall below 1100 trees after the winter plantings?                                                                                                |              |
| (iii) What will happen to the plantation into the future if conditions remain unchanged?                                                                                       |              |

**End of Question 5**

**End of Exam**

Question 1:

a)  $\log_{10} 12 - \log_{10} 2$

$= 0.77815125$

$= 0.778 \text{ (3dp)} \text{ ①}$

b)  $\sum_{n=2}^5 3n - 1$

$= 5 + 8 + 11 + 14$

$= 38 \text{ ①}$

or

$S_n = \frac{n}{2}(a + l)$

$= \frac{5}{2}(5 + 14)$

$= 38$

c)  $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$= \frac{|5x - 1 + -12y + 15|}{\sqrt{5^2 + (-12)^2}}$

$= \frac{|-5 - 36 + 15|}{\sqrt{169}}$

$= \frac{|-26|}{13}$

$= \frac{26}{13}$

$= 2 \text{ ①}$

d) i)

Domain: all  $x$  ①

Range:  $y \geq 0$  ①

ii)

Domain:  $x \leq 9$  ①

Range:  $y \geq 0$  ①

iii)

Domain:  $x \neq 0$  ①

Range:  $y = -1$  or  $1$  ①

e)  $d = 3$   $a = 2$

$T_{20} = a + (n-1)d$

$= 2 + 19 \times 3$

$= 59 \text{ ①}$

ii)  $S_n = \frac{20}{2}(2 + 59)$

$= 610 \text{ ①}$

or

$= \frac{20}{2}(2 \times 2 + (19)3)$

$= 610$

f) i)  $x = 9$  ①

ii)  $x = 2 + 9$  ①

## QUESTION 2

$$(a) \quad 3^x = 81$$
$$\log_3 81 = x$$
$$x = 4$$

$$(b) \quad \Delta = 16 - 4 \times 3 \times 2$$
$$= -8$$

$\Delta < 0 \therefore$  no real roots

$$(c) \quad 2x^2 - 5x + 8 = 2x^2 - 2x - 3x + 8$$
$$= 2x(x-1) - 3x + 8$$

$$(d) (i) \quad 9x^2$$

$$(ii) \quad \frac{(x+1) \times 2x - x^2 \times 1}{(x+1)^2}$$
$$= \frac{x^2 + 2x}{(x+1)^2}$$

$$(iii) \quad \sqrt{x^2+1} \times 1 + x \times \frac{1}{2}(x^2+1)^{-\frac{1}{2}} \times 2x$$
$$= \sqrt{x^2+1} + \frac{x^2}{\sqrt{x^2+1}}$$

(e)

$$x^2 + 6x - 16 = 0$$

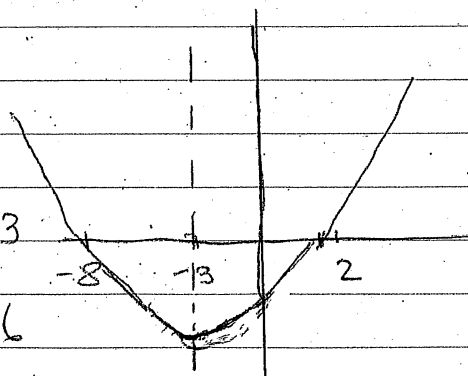
$$(x+8)(x-2) = 0$$

(i) axis of symmetry  $x = -3$

$$\text{min value } (-3)^2 + 6(-3) - 16$$
$$= -25$$

$$(ii) \quad f(x) \geq 0$$

when  $x \leq -8$  or  $x \geq 2$



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3 (a)  $f(x) = \frac{3x^3 + x - 1}{x}$

Is it even?  $f(x) = f(-x)$ .

$$f(-x) = \frac{3(-x)^3 + (-x) - 1}{(-x)}$$

$$= \frac{-3x^3 - x - 1}{-x}$$

$$= \frac{3x^3 + x + 1}{x} \quad \text{NO.}'$$

Is it odd?  $f(x) = -f(-x)$

$$-f(-x) = \frac{-3x^3 - x - 1}{x} \quad \text{NO.}'$$

$f(x)$  is neither ①

(b)  $\log_6 4 = 1.38 \Rightarrow 2\log_6 2 = 1.38$   
 $\log_6 2 = 0.69$  and  $\log_6 3 = 1.09$

$$\begin{aligned} \text{(i)} \log_6 12 &= \log_6 4 + \log_6 3 \\ &= 1.38 + 1.09 \\ &= 2.47 \quad \text{①} \end{aligned}$$

$$\text{(ii)} \log_6 2 - \log_6 3$$

$$= 0.69 - 1.09 = -0.4 \quad \text{①}$$

$$(c) \quad 2x^2 + 5x - 3 = (2x - 1)(x + 3)$$

$$4x^2 + 16x + 12 = 4(x^2 + 4x + 3) \\ = 4(x + 3)(x + 1)$$

$$f(x) = \frac{(2x - 1)(x + 3)}{4(x + 3)(x + 1)} = \frac{2x - 1}{4(x + 1)}$$

$$(i) \lim_{x \rightarrow 0} \frac{-1}{4} \quad (1)$$

$$(ii) \lim_{x \rightarrow -3} \frac{-7}{4x - 2} = \frac{-7}{-8} = \frac{7}{8} \quad (1)$$

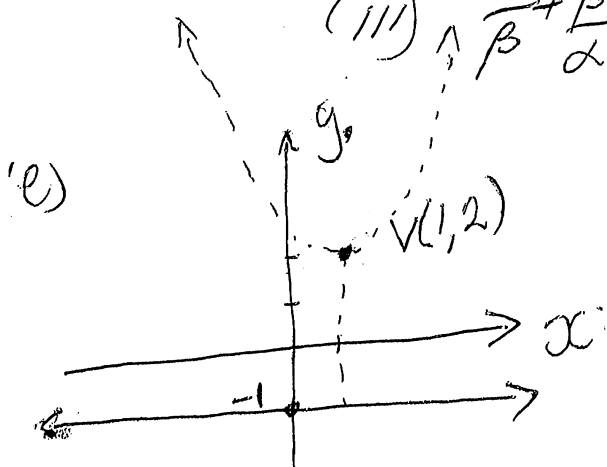
$$(iii) \lim_{x \rightarrow \infty} \frac{x^2 \left( 2 + \frac{5}{x} - \frac{3}{x^2} \right)}{x^2 \left( 4 + \frac{16}{x} + \frac{12}{x^2} \right)} \rightarrow \frac{1}{2} \quad (2)$$

$$(d) \quad a = 2 \\ b = 7 \\ c = 4$$

$$(i) \alpha + \beta = -\frac{b}{a} = -\frac{7}{2} \quad (1)$$

$$(ii) \alpha\beta = \frac{c}{a} = \frac{4}{2} = 2 \quad (1)$$

$$(iii) \frac{\frac{\alpha}{\beta} + \frac{\beta}{\alpha}}{\alpha\beta} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta} = \frac{\frac{49}{4} - 4}{2} \\ = \frac{25}{8} \quad (2)$$



$$a = 3$$

$$(x - h)^2 = 4a(y - k)$$

$$(x - 1)^2 = 12(y - 2) \quad (2)$$

# Question 4

(a)  $200 + 150 + 112\frac{1}{2} + \dots$

(i) limiting sum

$$S = \frac{a}{1-r}$$

$$= \frac{200}{1-\frac{3}{4}}$$

$$= \$800$$

(ii)  $S_{20} = \frac{200 \left[ \left( \frac{3}{4} \right)^{20} - 1 \right]}{\frac{3}{4} - 1}$

$$= \$797.46$$

$$S - S_{20} = \$2.54$$

b)  $f(x) = x^2 - 6x$

2 (i)  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{[(x+h)^2 - 6(x+h)] - [x^2 - 6x]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 + 2xh - 6h}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h - 6$$

$$= 2x - 6$$

(ii)

$$LHS = 2x f'(x) - f(2x)$$

$$= 2x(2x-6) - (4x^2 - 12x)$$

$$= 4x^2 - 12x - 4x^2 + 12x$$

$$= 0$$

$$= RHS$$

(c)  $M_{OP} \times M_{PA} = -1$

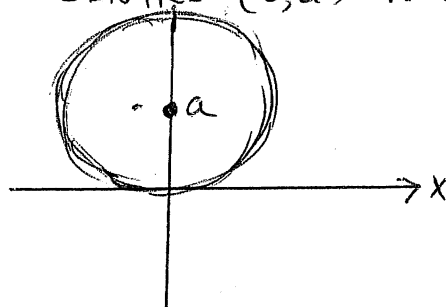
$$\Rightarrow \frac{y}{x} \times \frac{y-2a}{x} = -1$$

$$y^2 - 2ay + x^2 = 0$$

$$x^2 + y^2 - 2ay + a^2 = a^2$$

$$x^2 + (y-a)^2 = a^2$$

CIRCLE CENTRE  $(0, a)$  radius  $a$



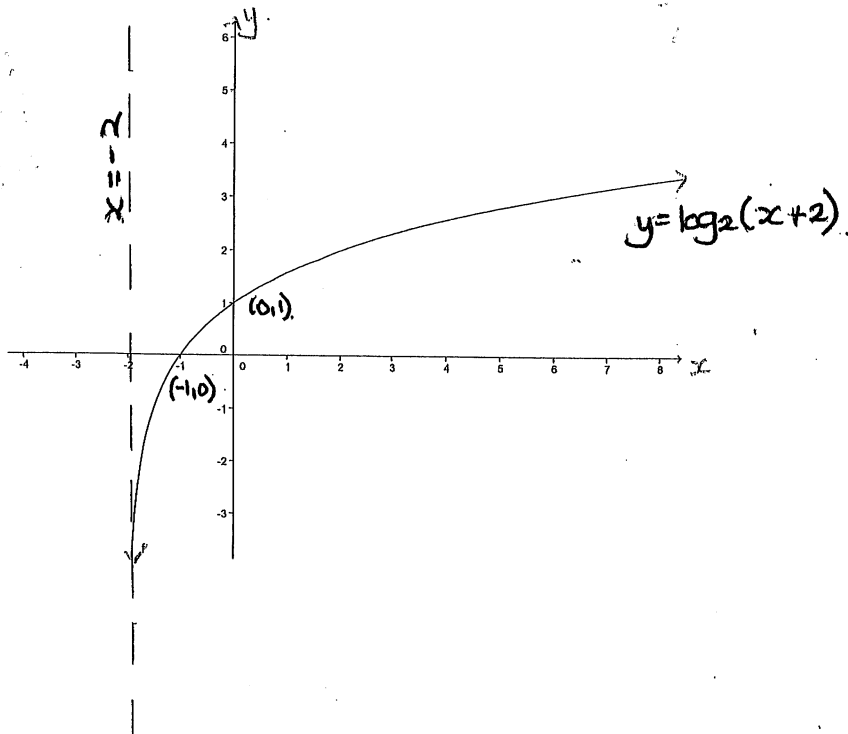
(d)

(i)  $\log_b c = \log_b a^2 = 2 \log_b a$   
 $= 2p$

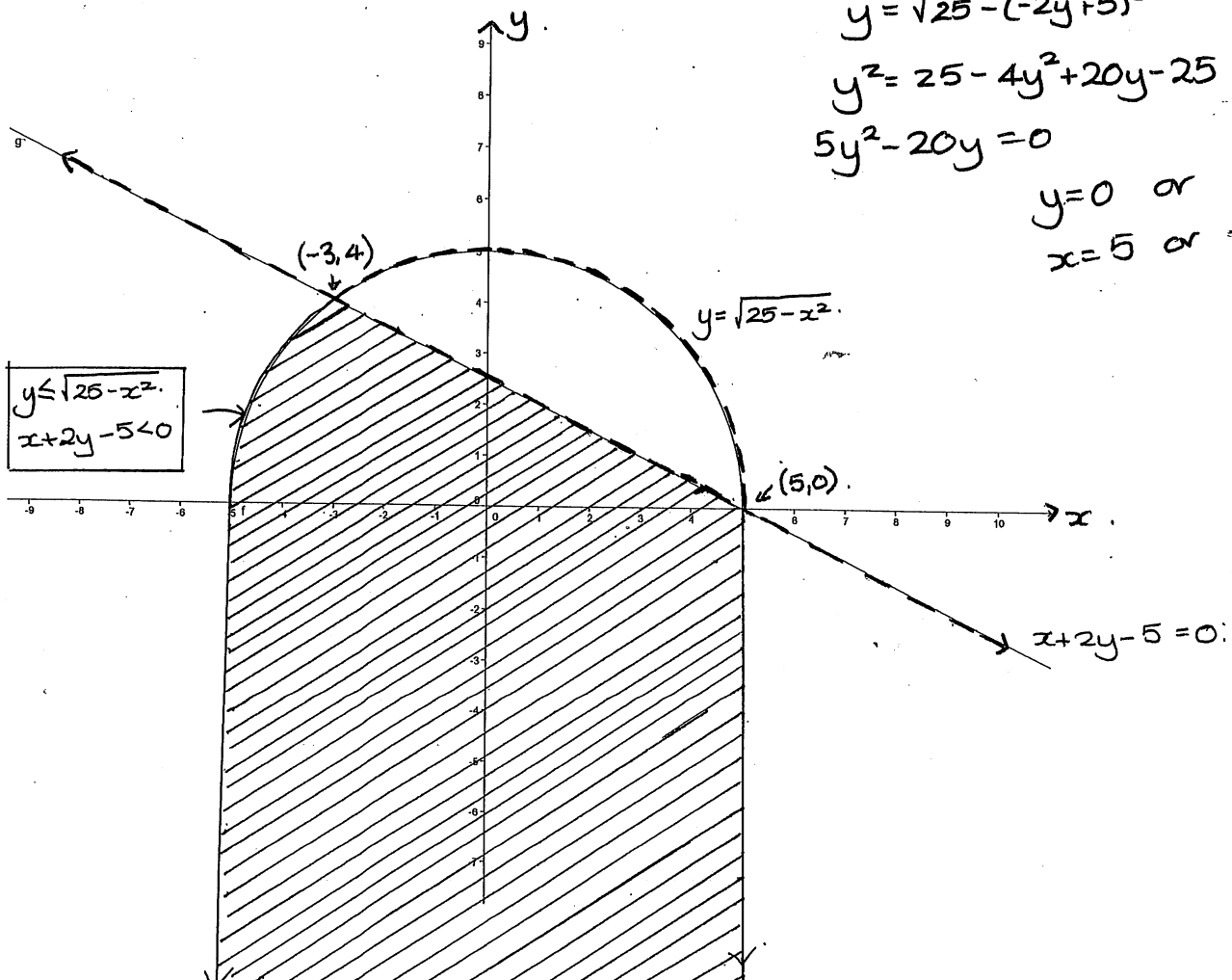
(ii)  $\log_c b = \frac{\log_b b}{\log_b c}$

$$= \frac{1}{2p}$$

QUESTION FIVE (a)



QUESTION FIVE (b)



$$y = \sqrt{25 - x^2} \quad x = -2y + 5$$

$$y = \sqrt{25 - (-2y + 5)^2}$$

$$y^2 = 25 - 4y^2 + 20y - 25$$

$$5y^2 - 20y = 0$$

$$y = 0 \text{ or } 4$$

$$x = 5 \text{ or } -3$$

$$y \leq \sqrt{25 - x^2}$$

$$x + 2y - 5 < 0$$

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### QUESTION 5

(i) 1990 =  $A_0 = 1200$  trees.

$$\begin{aligned} 1989 = A_1 &= (A_0 - 100) + \frac{(A_0 - 100)}{90} \times 10 \\ &= \frac{10}{9}(A_0 - 100) \\ &= \frac{10}{9}(1200 - 100) \end{aligned}$$

$$\begin{aligned} 1988 = A_2 &= \frac{10}{9}(A_1 - 100) \\ &= \frac{10}{9} \left[ \frac{10}{9}(1200 - 100) - 100 \right] \\ &= \left(\frac{10}{9}\right)^2(1200) - \left(\frac{10}{9}\right)^2(100) - \left(\frac{10}{9}\right)(100) \end{aligned}$$

$$\begin{aligned} 1987 = A_3 &= \frac{10}{9}(A_2 - 100) \\ &= \frac{10}{9} \left[ \left(\frac{10}{9}\right)^2(1200) - \left(\frac{10}{9}\right)^2(100) - \left(\frac{10}{9}\right)(100) - 100 \right] \\ &= \left(\frac{10}{9}\right)^3(1200) - \left(\frac{10}{9}\right)^3(100) - \left(\frac{10}{9}\right)^2(100) - \frac{10}{9}(100) \end{aligned}$$

$$A_n = \left(\frac{10}{9}\right)^n(1200) - (100) \left[ \left(\frac{10}{9}\right)^n + \left(\frac{10}{9}\right)^{n-1} + \left(\frac{10}{9}\right)^{n-2} + \dots + \left(\frac{10}{9}\right)^1 \right]$$

$$\begin{aligned} * 1980 = A_{10} &= \left(\frac{10}{9}\right)^{10}(1200) - 100 \sum_{r=1}^{10} \left(\frac{10}{9}\right)^r \\ &= \left(\frac{10}{9}\right)^{10}(1200) - 100 \left[ \frac{\left(\frac{10}{9}\right) \left( \frac{10}{9}^{10} - 1 \right)}{\frac{10}{9} - 1} \right] \\ &= 3441.5664 - 100(18.6797) \\ &= 1573.594398 \end{aligned}$$

In 1980 there were approx 1574 trees.

(ii) 1990 =  $A_0 = 1200$ .

$$\begin{aligned} A_1 = 1991 &= A_0 - (0.1)A_0 + 100 \\ &= 0.9(A_0) + 100 = 0.9(1200) + 100 \end{aligned}$$

$$\begin{aligned} A_2 = 1992 &= A_1(0.9) + 100 \\ &= (0.9)[(0.9)(1200) + 100] + 100 \\ &= (0.9)^2(1200) + (0.9)(100) + 100 \end{aligned}$$

$$\begin{aligned} A_3 = 1993 &= A_2(0.9) + 100 \\ &= (0.9)^3(1200) + (0.9)^2(100) + (0.9)(100) + 100 \end{aligned}$$

AAA

QUESTION 5 (cont)

$$\begin{aligned}
 \text{ii) } A_n &= (0.9)^n (1200) + (0.9)^{n-1} (100) + \dots + (0.9)(100) + 100 \\
 &= (0.9)^n (1200) + (100)(1 + 0.9 + \dots + 0.9^{n-1}) \\
 &= (0.9)^n (1200) + 100 \left( \frac{1(1 - 0.9^n)}{0.1} \right)
 \end{aligned}$$

The plantation will fall below 1100 trees when  
 $A_n < 1100$ .

$$0.9^n (1200) + 1000(1 - 0.9^n) < 1100$$

$$0.9^n (1200) + 1000 - (0.9)^n (1000) < 1100$$

$$200(0.9)^n + 1000 \leq 1100$$

$$200(0.9)^n \leq 100$$

$$(0.9)^n < 1/2$$

$$n(\log 0.9) < \log(0.5)$$

$$n > \frac{\log(0.5)}{\log(0.9)}$$

$$\log 0.9 < 1$$

$$n > 6.5788$$

After 7 years the plantation will fall below 1100 trees ie. after 1997 winter plantings

QUESTION 5 (iii).

If conditions remain unchanged then the plantation will tend to a limit.

as  $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \left[ (0.9)^n (1200) + 100 (1 + 0.9 + 0.9^2 + \dots + 0.9^{n-1}) \right]$$

$$= \lim_{n \rightarrow \infty} \left[ 0.9^n (1200) \right] + 100 \lim_{n \rightarrow \infty} (1 + 0.9 + 0.9^2 + \dots + 0.9^{n-1})$$

$$= 0 \times 1200 + 100 \left[ S_{\infty} \right] \quad S_{\infty} = \frac{a}{1-r}$$

$$= 0 + 100 \left[ \frac{1}{1-0.9} \right]$$

$$= 1000.$$

The number of trees will tend to a limit of 1000 into the future if conditions remain unchanged.